HOTEL MANAGEMENT SYSTEM

Report 1: Project Description

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1. Business Idea

The Hotel Management System is an information system proposed to help a hotel organise guest records, room information, reservations, payments, and guest check-in/check-out activities.

2. Problem Statement

When hotel activities are handled manually, staff may have difficulty tracking room availability, reservation dates, guest details, and payment status. This can lead to duplicated bookings, misplaced records, and slow service. A structured system model can clarify the information the hotel needs to manage.

3. Purpose and Objectives

The purpose is to model the main information and business rules of a hotel management system. Objectives are to maintain guest details, represent rooms and their availability, record reservations, track payments, and represent each guest stay.

4. Project Scope — Included

The proposed design covers guest registration and details; room number, type, price, and status; reservation creation, cancellation, and total calculation; payment records; receptionist activities; and check-in/check-out records.

5. Project Scope — Excluded

This assignment models the system using UML. It does not require a fully working application. Payroll, supplier inventory, restaurant operations, advanced accounting, and integration with external booking websites are excluded.

6. Main Classes and Responsibilities

Guest: stores guest identity and contact information. Reservation: stores booking dates and status, and calculates the total. Room: stores room details, price, and availability status. Payment: records amount, date, method, and payment status. Receptionist: represents staff who manage reservations and guest arrival/departure. Stay: records the actual check-in and check-out times.

7. Brief System Requirements

The model should allow a guest to have multiple reservations over time; each reservation is associated with one room; a reservation may have one or more payment records; a receptionist may manage many reservations; and a reservation may have zero or one Stay record until the guest checks in.

8. Assumptions

Each reservation in this simplified model is for one room. A reservation belongs to one guest and is assigned to one receptionist for tracking. Payment records can be split into multiple transactions. A Stay record is created when a reservation is checked in.